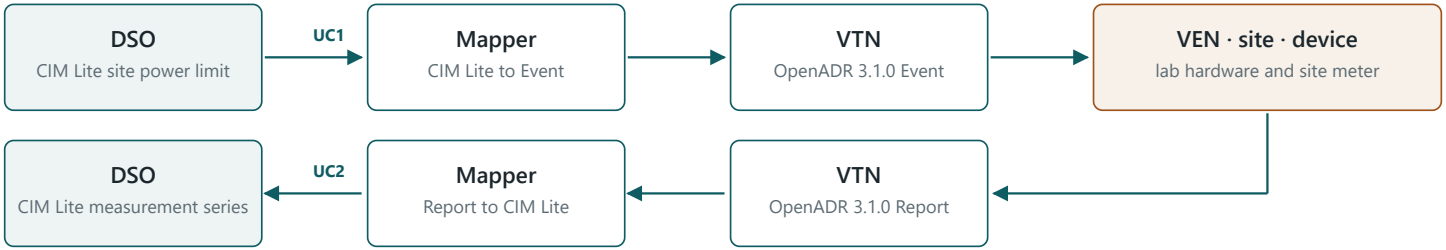
CIM–OpenADR Mapping Review

UC1 and UC2: mapping approach, open questions and proposals for the OpenADR Alliance

The supplied CEEDS/OpenADR documents bring resource identity, reference curves, operating envelopes and measurement quality into one assembly. We have used those concepts to build a small mapping to native OpenADR 3.1.0 Events and Reports. This review sets out the mapping choices, unresolved questions and proposed standards work.

The supplied `event` object is not a valid OpenADR 3.1.0 Event and cannot be used unchanged as an actionable native message (page 9). We propose an adapter from the agreed CIM semantics to native Events and Reports. We would like to confirm the assembly's intended role and exact source profile.



What is implemented

UC1 maps an absolute site operating envelope to an Event. UC2 maps a correlated measurement Report to a CIM Lite series. The example sends a **7.0 kW** import maximum and returns a **6.4 kW** interval mean. These are software examples, not hardware results.

What we propose to agree

Confirm the CIM roles, sign/time/unit rules and identity scope; separate profile agreements from standards gaps; then prepare focused Alliance proposals supported by examples and lab evidence.

CIM Lite is our proposed JSON application profile, not an IEC standard. VTN means the OpenADR server; VEN means the client/site controller. The December 2026 hardware test initially covers maximum import limits and interval-average import power. No new OpenADR payload is needed for that restricted scope.

Mapping the site operating envelope

The adapter reads the agreed CIM input semantics and creates a native Event request. The Event contains OpenADR intervals, targets and payloads; it does not require the VEN to interpret a nested MarketDocument.

INFORMATION	CIM LITE INPUT	OPENADR EVENT	RULE
Instruction identity	mRID **`<id>`** revisionNumber **`1`**	eventName **`cim-lite: OperatingEnvelope:<id>:1`**	The Event name carries the prefix, mRID and revision by profile convention. The VTN assigns the Event id. CL-09 · R-ID-2
Connection point	codingScheme **`A10`** value **`<EAN>`**	target **`cp:A10:<EAN>`**	Profile convention: cp:<scheme>:<value>. The same string is the Report target and resourceName. CL-02 · R-ID-1
Start time	start **`2026-10-16T16:00Z`**	start **`2026-10-16T16:00:00Z`**	Same instant: the ESMP minute timestamp becomes RFC 3339 UTC. CL-05 · R-TIME-1
Interval length	resolution **`PT1H`** curveType **`A01`**	duration **`PT1H`**	Fixed-size blocks (A01): the resolution becomes the default interval duration. CL-05 · R-TIME-3
Interval number	position **`1`**	id **`1`**	Profile convention: the point position becomes the interval id. OpenADR ids are not sequence numbers. CL-05 · R-ID-3
Limit type	limitNature **`hardLimit`** flowConvention **`load`**	payload type **`IMPORT_CAPACITY_LIMIT`**	A hard limit on import-positive power becomes a maximum import. Setpoints and relative values are rejected. CL-03 · CL-08 · CL-10 · R-ENV-1
Limit value	value **`7`**	value **`7.0`**	After unit conversion: 7 KWT = 7.0 kW. CL-03 · R-ENV-4
Unit	unit **`KWT`**	units **`KW`**	KWT becomes KW (WTT ×0.001, MAW ×1000). W with a multiplier is also accepted; Wh is energy and is rejected for a power limit. CL-07 · R-UNIT-1
Export limit	lowerLimit *absent*	not sent	No lower bound, so no EXPORT_CAPACITY_LIMIT is sent. Absent is not zero. CL-04 · R-ENV-3
Measurement request	mapper option	payloadType **`DEMAND`** readingType **`MEAN`**	The Event asks the VEN for the mean power in kW per Event interval, for the same target. CL-11 · R-OBS-1

Table identifiers are shortened. Example target: `cp:A10:871999999999999990`. Rule and CL identifiers refer to the project mapping matrix [S9]. Sender identity comes from configuration; the VTN assigns its own Event id and metadata. Revisions require a persisted CIM-id/revision ↔ Event-id mapping.

Mapping the measurements back to CIM

The Event requests `DEMAND`, reading type `MEAN`, unit `KW`. The adapter checks the Event id, target identifier, interval id and exact timing before returning a CIM Lite measurement series.

INFORMATION	OPENADR REPORT	CIM LITE OUTPUT	RULE
Event link	eventID **`evt-20261016-0001`**	instructionRef **`‹id›`**	The Report's eventID must equal the Event id. The CIM mRID is read back from the Event name. CL-15 · R-OBS-5
Connection point	resourceName **`cp:A10:‹EAN›`**	codingScheme **`A10`** value **`‹EAN›`**	Must be one of the Event targets; otherwise the Report is rejected. CL-02 · R-ID-1
Measurement type	payloadType **`DEMAND`** readingType **`MEAN`**	measurand **`ActivePower`** aggregation **`average`**	DEMAND with MEAN is the interval-mean active power. It describes the averaging, not how the meter measured it. CL-11 · R-OBS-4
Interval	start **`2026-10-16T16:00:00Z`** duration **`PT1H`**	start **`2026-10-16T16:00Z`** resolution **`PT1H`**	The interval id and its exact times must match the Event interval; otherwise the Report is rejected. CL-05 · R-OBS-5
Measured value	value **`6.4`**	value **`6.4`**	The interval mean is copied unchanged. CL-11 · R-OBS-6
Unit	units **`KW`**	unit **`KWT`**	DEMAND must be in kW; W is rejected by the profile. Written as KWT. CL-07 · R-UNIT-3
Quality	DATA_QUALITY absent	quality **`A04`**	Missing DATA_QUALITY maps to ENTSO-E A04 ('as provided'), with no assertion of validity. CL-13 · R-OBS-3
Evidence grade	profile default	evidenceGrade **`operationalTelemetry`**	OpenADR cannot mark data as settlement-grade, so it is recorded as operational telemetry. CL-16 · G-07

Metering alternative

IEC 61968-9 ReadingType can explicitly identify average power, direction and units. This is verified source evidence; a 61968-9 adapter is not implemented. Its interval timestamp denotes the end: 17:00 for a one-hour reading means [16:00, 17:00). ESMP positions start at 1; RegularTimePoint uses 0 at schedule start. [S3, S4]

What the result establishes

The example mean is 6.4 kW against a 7.0 kW maximum. That comparison does not prove instantaneous enforcement or command execution. Device logs, an independent meter, agreed tolerances and response-time criteria are needed for the hardware test. Reports remain operational telemetry.

CIM semantics and profile agreements

These items concern the source profile or the binding between systems. They should be reviewed with the CIM modelling team; an OpenADR enumeration addition cannot resolve a missing source-side meaning.

ITEM	FINDING	SUGGESTED ROUTE
G-02 · Release and enumeration differences	The supplied model, ESMP v5 and IEC 62746-4 use different schedule-kind names. A CIM100 namespace alone does not identify the exact release or profile.	Agree the UML/profile release, generator version and code-list version. Our profile uses load/generation and explicit aliases for consumption/production. Relative kinds remain separate. [S2, S3, S6]
G-03 · Physical connection-point identity	IEC 62746-4 identifies a RegisteredResource but does not provide the connection-point class needed by this mapping. A resource may cover a different physical scope from the site meter.	Use an explicit resource-to-connection-point registry. MarketEvaluationPoint.mRID is a verified ESMP anchor, but choosing a physical, non-aggregated point remains a profile decision. The intended boundary remains to be confirmed. [S2–S4]
G-04 · Envelope role and limit meaning	A curve value alone does not establish whether it is a maximum, a mandatory minimum, a setpoint or an advisory value. Our inspected 62746-4 profile has no dedicated envelope-role/nature fields.	Agree these meanings in the CIM profile and carry an explicit limit-nature assertion in the adapter configuration. Seek IEC profile clarification where needed. OpenADR maximum-capacity payloads already express a site maximum. [S2, S3]
G-11 · Meaning of an OpenADR target	Targets are opaque access-control strings. A matching identifier does not establish that the VEN controls the correct electrical connection.	Use cp:<scheme>:<value>, provision access and verify the registry against the site meter. A standard connection-point resource attribute is an open candidate under E-10. [S1]
G-15 · Units in the related EDDIE assembly	The reviewed EDDIE RECMMOE assembly omits an explicit unit for its min/max quantities. The INSIEME material does provide a series measurement unit; these are different artifacts.	Require an explicit unit, including the multiplier where applicable. This is a source-profile remedy, not a new OpenADR feature. A variable name or sample magnitude is insufficient to establish kW. [S6, S7]
G-17 · Quality on IEC 62746-4 curve values	The reviewed ReferenceEnergyCurve profile lacks a per-value measurement-quality field. Instruction status and MarketDER status do not qualify observations.	Retain quality in CIM Lite or use an appropriate metering profile. Ask IEC for an explicit quality channel if this profile is to carry observations. E-06 improves OpenADR vocabulary but cannot fix missing CIM-side quality. [S2, S4]
G-18 · Electrical quantities in IEC 62746-4	The reviewed measurement channels use W/Wh/kg; they do not supply the V/A/var/VA vocabulary needed for broader grid telemetry.	Use a suitable CIM measurement profile or agreed extension. OpenADR snapshots use READING with VOLTS / AMPS / KVAR for voltage, current or reactive power. Frequency and phase need E-11. [S1, S2, S4]

Control behaviour and instruction identity

Keep site enforcement, Event lifecycle and loss-of-connection behaviour explicit. A valid message alone does not establish that the correct device acted on the correct site limit.

ITEM	FINDING	SUGGESTED ROUTE
G-05 · Site limits and device control	IMPORT_CAPACITY_LIMIT and EXPORT_CAPACITY_LIMIT describe the whole site. Sending one to a device does not automatically change its physical scope.	Keep UC1 at the site boundary and let the site controller allocate the limit. A later device profile must define allocation, uncontrolled load and measurement scope; a setpoint is not an equivalent limit. [S1]
G-10 · External instruction reference	eventID links a Report to an OpenADR Event, but there is no standard field for the CIM instruction identity/revision. Event and Report have no generic attributes array.	Today: preserve the identity in eventName and a mapping registry. E-09 proposes a string EXTERNAL_REFERENCE payload for Event and Report. Define its scope, revision and multi-reference encoding before standardising it. [S1]
G-13 · Examples that disagree with the schema	The Guide's date-only 'start now' sentinel fails date-time validation. Other examples have missing descriptor objectType, inconsistent names/units, and ambiguous reading-type guidance.	Use explicit UTC timestamps and validate against the pinned OpenAPI plus payload enumeration schemas. Submit E-12 as an editorial correction package; it is not a new control payload. [S1]
G-19 · Reachability, failsafe and device response	An HTTP success response confirms VTN object acceptance, not device execution. The mapping has no implemented interoperable heartbeat/failsafe/device-write acknowledgement contract.	E-03: heartbeat Report and failsafe Program settings, with a defined local state machine. E-04: an Event-linked device-response Report. Successful polling can establish VTN reachability; push clients need an observable signal. [S1, S5]
G-20 · Minimum-capacity bands	Native maximum limits cannot require non-zero import/export. This minimum-band gap remains open; schedule fallback already has a profile solution.	E-01 proposes nonnegative minimum-capacity payloads with explicit enabled-direction semantics. Existing fallback uses longer, lower-priority Events (E-02). Communication timeout remains a separate E-03 issue. [S1, S5]

Minimum-band semantics (E-01)

For an import-positive signed envelope [l,u] containing zero, send IMPORT=u and EXPORT=−l for the bounds present. An absent bound is not zero. Forced import (l>0) or forced export (u<0) is currently rejected. Two enabled directional bands may form a disjoint feasible set; minimum payloads need explicit direction-enable rules.

Obligation and recovery (E-13)

Hard limit versus operating point is separate from binding versus advisory. BINDING_EVENTS means immutable Events, not legal/operational obligation. Agree this locally for the first test. Schedule fallback must not be confused with a timeout that overrides an otherwise active instruction.

Observation direction, quality and timing

The first UC2 contract is deliberately narrow: interval-average active power at the site boundary, with import-only reporting. Broader telemetry needs explicit channel and quality semantics.

ITEM	FINDING	SUGGESTED ROUTE
G-06 · Interval export power and energy	DEMAND and USAGE have nonnegative values; READING is instantaneous. A negative interval mean cannot be relabelled as READING without changing its meaning.	E-05 proposes EXPORT_DEMAND in KW and EXPORT_USAGE in KWH, both nonnegative. A future signed NET_DEMAND is an alternative design, not part of today's implementation. Agree directional/net aggregation and avoid double counting. [S1, S4]
G-12 · Quality vocabulary and granularity	Several ENTSO-E quality codes lack an OpenADR value. BAD has no direct ENTSO-E code. IEC metering quality and Quality61850 are further, distinct vocabularies.	E-06 proposes ADJUSTED, INCOMPLETE, CALCULATED, TEMPORARY and OUT_OF_RANGE. Keep source provenance and conservative crosswalks. Define which measurement each quality value qualifies; adding labels does not settle that association. [S1, S4, S8]
G-14 · Energy provenance per interval	USAGE's descriptor-level readingType, default DIRECT_READ, can assert one provenance for a whole series containing measured and derived intervals.	Keep energy reporting outside the current minimal UC2 profile. Before adding it, agree readingType and interval quality/provenance rules; per-interval DATA_QUALITY already exists. Test those rules before proposing an additional carrier. [S1]
G-21 · Snapshots, interval means and phase	TR 62746-2 monitoring includes latest values and per-phase quantities. UC2 currently carries interval-average active power; that is a different temporal contract.	Use READING for actual snapshots and DEMAND/MEAN for interval power. E-11 proposes HZ and an unambiguous phase encoding. Fix sampling, aggregation, timestamps and channel-quality association in the profile. [S1, S4, S5]

Rules to keep even after extending the vocabulary

- A missing measurement uses a zero placeholder with DATA_QUALITY=MISSING in the current profile. The receiver must ignore the placeholder; it is not a measured zero. Missing/estimated/bad values do not establish compliance.
- No DATA_QUALITY maps to ENTSO-E A04 “as provided”, without an assertion of validity. Metering DataValid (x.0.0) does not mean formally Validated (x.0.1). Do not strengthen the source's quality claim.
- Agree net versus import-register power, integration/sampling, interval coverage, clock offsets and meter orientation. A nonnegative net mean may conceal export or limit excursions within the interval.
- The example requests one historical Report at the end of the last interval. That is not live monitoring; agree a separate reporting cadence and independent measurement record for the lab.

Flexibility, dispatch and commercial exchange

The following gaps remain part of the mapping review, although they are not prerequisites for the initial maximum-limit/import-observation test. Existing value carriers should be reused where their semantics fit.

ITEM	FINDING	SUGGESTED ROUTE
G-07 · Settlement evidence and corrections	Operational Reports do not carry the evidence grade, correction history and audit semantics needed to establish settlement-grade measurements.	Keep settlement in the metering/settlement chain, linked to the operational instruction. A flag alone would not create settlement evidence. This is outside the first UC1/UC2 test. [S1, S4]
G-08 · Reference-curve role and baseline method	A forecast, a contractual reference curve and a settlement baseline may contain the same numbers but have different roles. Native value carriers do not identify that full contract or methodology.	Assert curveRole explicitly. Use DEMAND/FORECAST for supported nonnegative power forecasts and USAGE_FORECAST for energy. Keep methodology, curve-instance identity, revision and contractual role in the flexibility interface. [S1, S2]
G-09 · Two-sided relative flexibility band	A min/max load reduction band depends on a specified reference curve. It is not the same object as an absolute site operating envelope.	Keep it in the flexibility interface until reference identity/version, time alignment, units and obligation are fixed. Then an agreed transform may use P in [P_ref − r_max, P_ref − r_min]. This is not a current native carrier. [S2, S6]
G-16 · Offers and commercial lifecycle	Offers, acceptance, commitments and settlement require a commercial process beyond native operational Events. Existing PRICE/EXPORT_PRICE payloads do not provide that entire process.	Use a separate flexibility/commercial interface, drawing on appropriate CIM profiles, with an explicit handover reference into OpenADR. We are not proposing to rebuild the commercial lifecycle inside an Event. [S1, S2]
G-22 · Signed and directional dispatch	Existing consumption setpoints do not cover absolute export and all relative directions in the reviewed CIM instructions.	E-07 proposes export and directional-relative payloads. Specify consumption/generation direction, reference identity and timing before claiming complete coverage. The current candidate names do not by themselves resolve every generation-relative case. [S1, S2]
G-23 · Release of control	A CIM RELEASE instruction is not explicitly represented within the limit/dispatch signal family. Ending an interval is not always a sufficiently explicit hand-back instruction.	E-08 proposes NORMAL_OPERATION=true. Define its scope and interaction with other active Events and subscriptions. ACTIVATE versus ADJUSTMENT remains a separate loss. [S1, S2]

Prices and forecasts are not wholly absent from OpenADR: PRICE, EXPORT_PRICE, DEMAND/FORECAST and USAGE_FORECAST are existing carriers. Their availability does not make offers, contractual baselines, signed forecast bands or settlement workflows equivalent to an operational Event.

What we could take to the OpenADR Alliance

Proposed additions are candidates for Alliance discussion, with working names. E-12 corrects the published material, and E-10/E-13 need a decision. Fallback is handled by an existing profile rule and is not listed as a standards gap.

REQUEST / TOPIC	DECISION	DEFINITION OR WORK STILL NEEDED
E-01 · Minimum-capacity payloads	Proposed addition	IMPORT_CAPACITY_MINIMUM / EXPORT_CAPACITY_MINIMUM. Define enabled directions, zero and obligation; disjoint envelopes also need a richer CIM model.
E-03 · Heartbeat and failsafe	Proposed addition	HEARTBEAT Report; timeout/duration/import/export failsafe Program settings. Fix payload value type, event linkage and recovery behaviour.
E-04 · Device response	Proposed addition	EVENT_RESPONSE Report: ACCEPTED / REJECTED / NOT_SUFFICIENT; optional reason. Define response timing and what accepted/executed means.
E-05 · Export observations	Proposed addition	EXPORT_DEMAND (KW) and EXPORT_USAGE (KWH), nonnegative. Preserve interval and aggregation semantics.
E-06 · Measurement quality	Proposed addition	Extend DATA_QUALITY vocabulary, with conservative source crosswalks and an explicit association to measured values.
E-07 · Directional dispatch	Proposed addition	New export and relative-direction payloads. Resolve generation-relative cases, reference role and binding force.
E-08 · Release of control	Proposed addition	NORMAL_OPERATION=true within the relevant control context. Define precedence and hand-back behaviour.
E-09 · External reference	Proposed addition	String EXTERNAL_REFERENCE payload on Event/Report; specify identity, revision, interval scope and encoding.
E-10 · Resource attributes	Open	Candidates: GRID_CONNECTION_POINT, ramp rates and MAX_RESPONSE_TIME. Agree units, scope and which capability is being declared.
E-11 · Frequency and phase	Proposed addition	Add HZ and either per-phase payload types or a PHASE companion. Resolve value/phase/quality pairing for multiple channels.
E-12 · Specification errata	Editorial correction	Correct dates, required fields, identifiers, payload/unit names and inconsistent reading-type guidance. Editorial corrections, not new payloads.
E-13 · Binding versus advisory	Open	Decide between a Program agreement and a per-Event signal. BINDING_EVENTS concerns immutability, not binding force.

Most candidates aim to add enumeration entries while retaining the 3.1.0 object shapes. Passing the permissive base schema does not make a private string a standard payload or assure vendor support. Each addition still needs semantics, units, cardinality, examples, validation rules and an agreed profile. [S1, S9]

Mapping the supplied CEEDS/OpenADR documents

As supplied, the `event` object in the 2 October 2026 CEEDS schema is **not a valid OpenADR 3.1.0 Event**: its identifier types and metadata do not match the standard, and it supplies no actionable native intervals, payloads or targets. It cannot be sent unchanged as the operational message. We propose an adapter from the CIM assembly to native Events and Reports.

The CEEDS draft exposed two important integration questions: schedule-kind differences across models and the need for minimum-capacity semantics. The domain concepts are a useful starting point; the binding needs correction.

OBSERVATION IN THE SUPPLIED DRAFT	NATIVE OPENADR 3.1.0 BINDING	SUGGESTED NEXT STEP
Event-like wrapper with a nested MarketDocument; no native actionable intervals/payloads/targets. Empty roots/Points and metadata-only Events are accepted by the supplied schema.	An Event has native intervals and payloads. A VEN needs their standard meaning to act.	Confirm whether the assembly is a semantic reference, an adapter input or an encapsulated exchange format. Our preference is an explicit adapter into native Events/Reports.
event.id/programID are integers; modifiedDateTime is used. Server metadata is required in the supplied schema.	Object ids are strings; native metadata uses modificationDateTime. The VTN provisions Event metadata.	Separate adapter input from Event request and server-response representations. programID must reference a provisioned Program.
Quantity roles rely on quantityDefinition/quantityQualifier and related min/max assembly roles.	IMPORT/EXPORT_CAPACITY_LIMIT have defined site-maximum meanings and nonnegative values.	Agree controlled role values and the exact source release. These names were not found in the inspected 62325-351:2016/-301:2018 editions; a later source may explain them.
Document, series and resource identifiers coexist.	Event id, Program id, target and Report resourceName have separate scopes.	Retain the distinctions and document their correlation. Confirm which source mRID identifies the instruction and which denotes the physical site.
The delivered assembly supplies no Report-side binding.	Event reportDescriptors request native Reports linked by eventID and interval identity/timing.	Review UC2's measurement definition, quality and correlation together; add a Report-side adapter contract.
The bundled standard/local code-list composition uses oneOf, with overlapping codes in the reviewed bundle.	A value matching both branches fails oneOf even if it is a legitimate code.	Confirm the dependency bundle/version. Use a deliberate union or disjoint local lists. This finding concerns that generated dependency, not ENTSO-E code lists generally.

Evidence: the supplied CEEDS schema, the local 7 October validation findings, and test_insieme_style_event_fails_wire_schema [S6, S9]. EDDIE transformer findings [S7] are a separate comparison; they do not establish how the INSIEME implementation behaves. An encapsulated document would need its own exchange contract.

Questions for profile review

1. **Source profile.** Which CIM UML release, profile package, generator and code lists underlie the supplied assembly? Can we trace the quantity-definition and min/max roles to their authoritative source?
2. **Envelope contract.** Are bounds absolute site net power, per-direction bands, or changes relative to a reference? Which roles are hard constraints, which are operating points, and which are binding or best effort?
3. **Physical scope and identity.** How should RegisteredResource, the measurement point and the electrical connection be related? Which identifiers must survive updates and appear in returned measurements?
4. **UC2 contract.** Which measurement profile best represents the intended observation: the current Quantity model, IEC 61968-9 metering, or IEC 62746-4 with an explicit agreement? Which quality and aggregation semantics must survive?
5. **Joint Alliance input.** Which definitions can we agree now and demonstrate in the lab? We suggest starting with device response/reachability, export/quality reporting and external references, while keeping minimum bands and broader dispatch clearly scoped. The errata package can be prepared separately.

Our suggested next step is to resolve the source-profile questions, agree a small set of positive and counterexample messages, and turn the confirmed OpenADR gaps into individual change requests. Each request should include the CIM meaning, a wire example, precedence/failure behaviour and an interoperability check.

Sources and status of this review

[S1] OpenADR 3.1.0: pinned openadr3.yaml, payload/reading-type/unit enumeration schemas, Definition and User Guide; release commit ddd2cc58589bdc31ba1dd19fbdf90ed7e10b31dc. Public specification.

[S2] IEC 62746-4:2024, full text and Annex C schemas. **[S3]** IEC 62325-351:2016; IEC 62325-301:2018; ENTSO-E ESMP UML v5. **[S4]** IEC 61968-9:2024, §5.3.3 and Annexes C/D; IEC 61968-11:2013; IEC 61970-301:2020+A1:2022. **[S5]** IEC TR 62746-2:2025, JWG3000–3006 (informative use cases, not an OpenADR conformance requirement).

[S6] The supplied CEEDS/OpenADR documents: generated assembly schema, annotations and two model diagrams, 2026-10-02, CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnveloppe; our local validation findings, 2026-10-07. **[S7]** EDDIE RECMMOE v1.12 and transformer fixture, reviewed separately. **[S8]** ENTSO-E code lists v96r1 (2026-08-04); supplied bundle refers to v91.

[S9] Project mapping matrix, semantic rules, comparison, gap register and extension requests (docs 04–07/10), plus the validated example inputs and mapper run. Open items retain their original G/E register identifiers; closed gaps are omitted. Fallback remains documented as an existing profile rule, rather than an Alliance change request. Licensed standards are paraphrased; their PDFs are not attached. Source editions differ and do not establish every later CIM release.

Implementation boundary. The examples pass schema, enumeration and applicable profile/correlation checks. The local envelope round trip preserves the model values with sender context supplied externally. This is not a hardware test, certification, or a claim that the complete CIM/OpenADR standards are losslessly interchangeable. Candidate payloads are not implemented. The current XML adapter writes envelope ReferenceEnergyCurve XML only; it reads ReferenceEnergyCurve, Instruction and Price XML. No 61968-9 meter adapter is claimed.